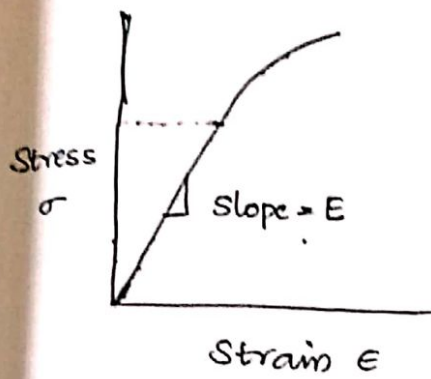
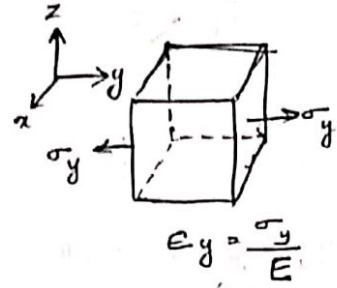
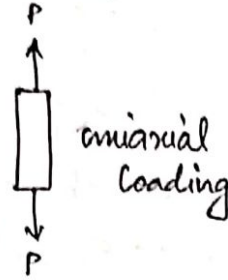


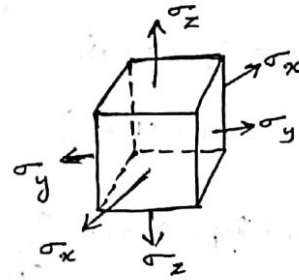


Tension test



$$\epsilon_x = -\nu \epsilon_y = -\nu \frac{\sigma_y}{E}$$

$$\epsilon_z = -\nu \frac{\sigma_x}{E}$$

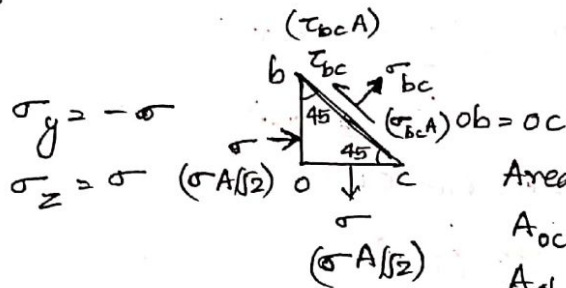
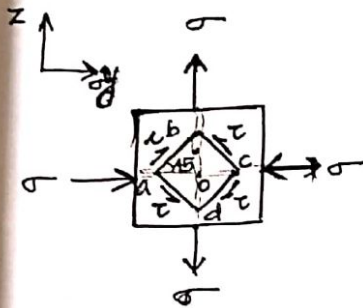


Poisson's ratio, $\nu = \frac{\text{Lateral strain}}{\text{Longitudinal strain}}$

$$\frac{\sigma_x}{E} - \frac{\nu}{E}(\sigma_y + \sigma_z) = \epsilon_x = \frac{-\nu \sigma_x}{E} + \frac{\sigma_x}{E} - \frac{\nu \sigma_z}{E}$$

$$\frac{\sigma_y}{E} - \frac{\nu}{E}(\sigma_x + \sigma_z) = \epsilon_y = \frac{\sigma_y}{E} - \frac{\nu \sigma_x}{E} - \frac{\nu \sigma_z}{E}$$

$$\frac{\sigma_z}{E} - \frac{\nu}{E}(\sigma_x + \sigma_y) = \epsilon_z = \frac{-\nu \sigma_x}{E} + \frac{\sigma_z}{E} - \frac{\nu \sigma_y}{E}$$

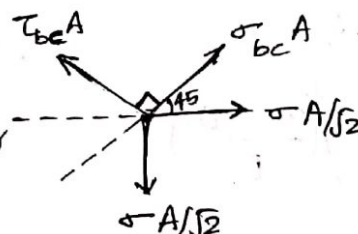


$$\text{Area } bc = A$$

$$A_{oc} = A \cos 45^\circ = A/\sqrt{2}$$

$$A_{ob} = A \sin 45^\circ = A/\sqrt{2}$$

No normal stress $\Rightarrow$ pure shear

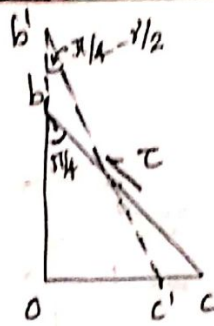


$$\sigma_{bc}A + \sigma A/\sqrt{2} \cos 45^\circ - \sigma A/\sqrt{2} \cos 45^\circ = 0$$

$$\sigma_{bc}A = 0 \Rightarrow \sigma_{bc} = 0$$

$$\tau_{bc}A - \sigma A/\sqrt{2} \cos 45^\circ - \sigma A/\sqrt{2} \cos 45^\circ = 0$$

$$\tau_{bc} = \sigma$$



$$\epsilon_y = \frac{\sigma_y}{E} - \frac{\nu}{E}(\sigma_x + \sigma_z)$$

$$= \frac{-\sigma}{E} - \frac{\nu}{E}\sigma$$

$$= -\frac{\sigma}{E}(1+\nu)$$

$$OC = OB = dl$$

Change in length $CC' = OC \epsilon_y = dl \epsilon_y$

$$OC' = OC + CC' = dl + dl \epsilon_y$$

$$= dl(1 + \epsilon_y)$$

$$\epsilon_z = \frac{\sigma_z}{E} - \frac{\nu}{E}(\sigma_x + \sigma_y) = \frac{\sigma}{E} - \frac{\nu}{E}(-\sigma)$$

$$= \frac{\sigma}{E}(1+\nu)$$

$$OB' = dl(1 + \epsilon_z)$$

$$\tan(\pi/4 - \gamma/2) = \frac{OC'}{OB'} = \frac{dl(1 + \epsilon_y)}{dl(1 + \epsilon_z)} = \frac{1 + \epsilon_y}{1 + \epsilon_z}$$

$$\frac{\tan \pi/4 - \tan \gamma/2}{1 + \tan \pi/4 \tan \gamma/2} = \frac{1 + \epsilon_y}{1 + \epsilon_z}$$

$$= \frac{1 - \frac{\sigma}{E}(1+\nu)}{1 + \frac{\sigma}{E}(1+\nu)}$$

$$\frac{1 - \gamma/2}{1 + \gamma/2} = \frac{1 - \frac{\sigma}{E}(1+\nu)}{1 + \frac{\sigma}{E}(1+\nu)}$$

Comparing LHS and RHS

$$\gamma = \frac{2\sigma}{E}(1+\nu)$$

$$\tau = \sigma$$

γ -Shear strain

$$\gamma = \frac{2\tau}{E}(1+\nu)$$

$$\tau \propto \gamma$$

Shear modulus / Modulus of rigidity $G = \frac{1}{2} \left(\frac{2(1+\nu)}{E} \right)$

$$G = \frac{E}{2(1+\nu)}$$

$$\gamma = \frac{\tau}{G}$$

Generalized Hooke's law

$$\gamma_{yz} = \frac{\tau_{yz}}{G}$$

$$\gamma_{xy} = \frac{\tau_{xy}}{G}$$

$$\gamma_{xz} = \frac{\tau_{xz}}{G}$$

$$\begin{aligned}\sigma_x &= (E\epsilon_x + \nu(E(\sigma_y + \sigma_z)))E \\ &= E\epsilon_x + \nu(E\epsilon_y + E\epsilon_z) \\ &= E(E\epsilon_x + \nu(E\epsilon_y + E\epsilon_z))\end{aligned}$$

$$\begin{aligned}\sigma_y &= E(E\epsilon_y + \nu(E\epsilon_x + E\epsilon_z)) \\ \sigma_z &= E(E\epsilon_z + \nu(E\epsilon_x + E\epsilon_y))\end{aligned}$$

$$\tau_{yz} = \gamma_{yz} G$$

$$\tau_{xz} = \gamma_{xz} G$$

$$\tau_{xy} = \gamma_{xy} G$$

OR

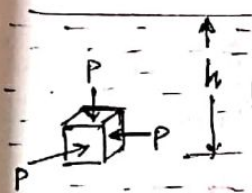
$$\epsilon_x = \frac{\sigma_x}{E} - \frac{\nu}{E}(\sigma_y + \sigma_z) \quad \text{--- ①}$$

$$\epsilon_y = \frac{\sigma_y}{E} - \frac{\nu}{E}(\sigma_x + \sigma_z)$$

$$\epsilon_z = \frac{\sigma_z}{E} - \frac{\nu}{E}(\sigma_x + \sigma_y)$$

Volume expansion, $e = \epsilon_x + \epsilon_y + \epsilon_z$

$$= \frac{\sigma_x + \sigma_y + \sigma_z}{E} - \frac{\nu 2(\sigma_x + \sigma_y + \sigma_z)}{E}$$



$$\sigma_x = \sigma_y = \sigma_z = -P$$

$$e = \frac{\sigma_x + \sigma_y + \sigma_z}{E} (1 - 2\nu) \quad \text{②}$$

$$= -3P \left(\frac{1 - 2\nu}{E} \right)$$

$$\frac{P}{e} = \frac{-E}{3(1 - 2\nu)} = K \quad \text{Bulk modulus}$$

$$\sigma_x = E\epsilon_x + \nu(\sigma_y + \sigma_z) \quad \text{--- ③}$$

From ③

$$\sigma_z + \sigma_y = \frac{eE}{1 - 2\nu} - \sigma_x$$

$$\begin{aligned}\sigma_x &= E\epsilon_x + \nu \left(\frac{eE}{1 - 2\nu} - \sigma_x \right) \\ &= E\epsilon_x + \frac{eE\nu}{1 - 2\nu} - \nu\sigma_x\end{aligned}$$

$$(1+\nu)\sigma_x = \frac{\nu E}{1-2\nu} e + E\epsilon_x$$

$$\sigma_x = \frac{\nu E}{(1+\nu)(1-2\nu)} e + \frac{E}{1+\nu} \epsilon_x$$

$$\sigma_x = \lambda e + 2G\epsilon_x$$

$$G = \frac{E}{2(1+\nu)}$$

$$\sigma_y = \lambda e + 2G\epsilon_y$$

$$\lambda = \frac{\nu E}{(1-2\nu)(1+\nu)}$$

$$\sigma_z = \lambda e + 2G\epsilon_z$$

Lame's constant

$$\gamma_{xy} = \frac{\tau_{xy}}{G}$$

$$\tau_{xy} = G\gamma_{xy}$$

$$\tau_{xz} = G\gamma_{xz}$$

$$\tau_{yz} = G\gamma_{yz}$$

Stress Matrix $\sigma = \begin{bmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{bmatrix} = \tau_{ij}$

Strain Matrix $\epsilon = \begin{bmatrix} \epsilon_x & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{yx} & \epsilon_y & \epsilon_{yz} \\ \epsilon_{zx} & \epsilon_{zy} & \epsilon_z \end{bmatrix} = \epsilon_{ij}$

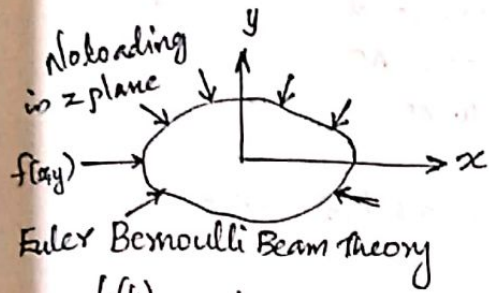
$$\epsilon_{xy} = \gamma_{xy}/2$$

$$\epsilon_{yz} = \gamma_{yz}/2$$

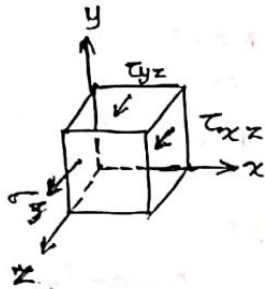
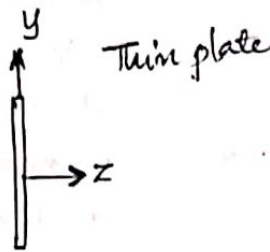
$$\epsilon_{zx} = \gamma_{zx}/2$$

$$\tau_{ii} = \tau_{xx} + \tau_{yy} + \tau_{zz}$$

Plane Stress



$$\frac{b(h)}{l} < \frac{1}{10}$$



$$\left. \begin{aligned} \sigma_y = \tau_{yz} = \tau_{xz} = 0 \\ \sigma_x = \sigma_y = \tau_{xy} = f(x, y) \end{aligned} \right\} \text{Plane Stress}$$

$$\epsilon_z = \frac{\sigma_z}{E} - \frac{\nu}{E} (\sigma_y + \sigma_x) = \frac{\sigma_z}{E} - \frac{\nu}{E} \sigma_y$$

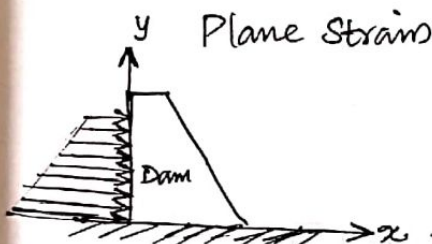
$$\epsilon_y = \frac{\sigma_y}{E} - \frac{\nu}{E} (\sigma_x + \sigma_z) = \frac{\sigma_y}{E} - \frac{\nu}{E} \sigma_x$$

$$\gamma_{xy} = \frac{\tau_{xy}}{G}$$

$$\gamma_{yz} = \frac{\tau_{yz}}{G} = 0$$

$$\gamma_{xz} = \frac{\tau_{xz}}{G} = 0$$

$$\epsilon_z = \frac{\sigma_z}{E} - \frac{\nu}{E} (\sigma_x + \sigma_y) = -\frac{\nu}{E} (\sigma_x + \sigma_y)$$



same perpendicular loading at every point

$$u, v = f(x, y) \\ \Rightarrow \frac{\partial f}{\partial z} = 0$$

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \Rightarrow \tau_{xy} = \frac{\gamma_{xy}}{G}$$

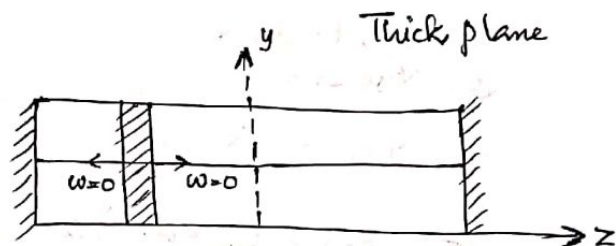
$$\gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} = 0 \\ \Rightarrow \tau_{yz} = 0$$

$$\gamma_{xz} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} = 0 \\ \Rightarrow \tau_{xz} = 0$$

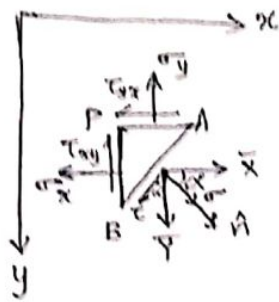
$$\epsilon_x = \frac{\partial u}{\partial x} \\ \epsilon_y = \frac{\partial v}{\partial y} \\ \epsilon_z = \frac{\partial w}{\partial z} = 0$$

$$\epsilon_z = \frac{1}{E} (\sigma_z - \nu (\sigma_x + \sigma_y)) = 0$$

$$\sigma_z = \nu (\sigma_x + \sigma_y)$$



displacement is z-direction, $w=0$ in every sections



α is measured in counter clockwise direction to x

$\bar{X}, \bar{Y}$ components of stresses on face AB

$\hat{n}$ = normal to AB

$\hat{n} = l, m, n$

$= \cos(\hat{n}, x), \cos(\hat{n}, y), \cos(\hat{n}, z)$

$l = \cos \alpha$

$m = \sin \alpha$

$n = \cos 90^\circ = 0$

normal and z axis

Area of AB = $A_{AB} = A$

Area of PA = $A_{PA} = A \cos(90^\circ - \alpha)$
= $A m$

Area of PB = $A_{PB} = A \cos \alpha = A l$

$$\bar{X} A - \sigma_x A l - \tau_{yx} A m = 0$$

$$\bar{X} - \sigma_x l - \tau_{yx} m = 0$$

$$\bar{X} = \sigma_x l + \tau_{yx} m$$

→

$$\bar{Y} = \sigma_y m + \tau_{xy} l$$

Components of stress in any arbitrary plane

In 3D

$$\bar{X} = \sigma_x l + \tau_{yx} m + \tau_{xz} n$$

$$\bar{Y} = \sigma_y m + \tau_{xy} l + \tau_{yz} n$$

$$\bar{Z} = \sigma_z n + \tau_{xz} l + \tau_{yz} m$$

τ - Shear stress on plane AB

σ - Normal stress on plane AB

$$\sigma = \bar{X} \cos \alpha + \bar{Y} \sin \alpha$$

$$\tau = \bar{Y} \cos \alpha - \bar{X} \sin \alpha$$

$$\sigma = (\sigma_x \cos \alpha + \tau_{yx} \sin \alpha) \cos \alpha + (\sigma_y \sin \alpha + \tau_{xy} \cos \alpha) \sin \alpha$$

$$= \sigma_x \cos^2 \alpha + \tau_{yx} \sin \alpha \cos \alpha + \sigma_y \sin^2 \alpha + \tau_{xy} \cos \alpha \sin \alpha$$

$$\rightarrow \sigma = \sigma_x \cos^2 \alpha + \sigma_y \sin^2 \alpha + 2 \tau_{xy} \sin \alpha \cos \alpha \quad \text{---(A)}$$

$$\tau = (\sigma_y \sin \alpha + \tau_{xy} \cos \alpha) \cos \alpha - (\sigma_x \cos \alpha + \tau_{yx} \sin \alpha) \sin \alpha$$

$$= \sigma_y \sin \alpha \cos \alpha + \tau_{xy} \cos^2 \alpha - \sigma_x \cos \alpha \sin \alpha - \tau_{yx} \sin^2 \alpha$$

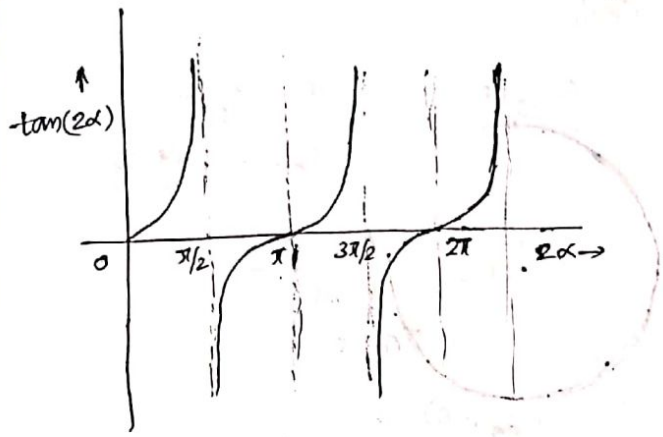
$$= (\sigma_y - \sigma_x) \sin \alpha \cos \alpha + \tau_{xy} \cos^2 \alpha - \tau_{yx} \sin^2 \alpha$$

$$\rightarrow \tau = (\sigma_y - \sigma_x) \sin \alpha \cos \alpha + \tau_{xy} (\cos^2 \alpha - \sin^2 \alpha) \quad \text{---(A')}$$

On a zero shear plane, $\tau = 0$

$$(\sigma_y - \sigma_x) \sin \alpha \cos \alpha + \tau_{xy} (\cos^2 \alpha - \sin^2 \alpha) = 0$$

$$\frac{\tau_{xy}}{\sigma_y - \sigma_x} = \frac{\sin \alpha \cos \alpha}{\sin^2 \alpha - \cos^2 \alpha} = \frac{1/2 \sin 2\alpha}{\cos 2\alpha} = \frac{1}{2} \tan(2\alpha)$$



We have two zero shear plane and they are perpendicular

(or. $\alpha_1, \alpha_2 = \alpha, \alpha + \pi/2$)

These are called principal planes.

The stress on principal planes are called principal stress

If x and y planes are the principal planes.

Then $\tau_{xy} = \tau_{yx} = 0$

Sub in A and A'

$$\sigma = \sigma_x \cos^2 \alpha + \sigma_y \sin^2 \alpha \quad \text{--- B*}$$

$$\tau = (\sigma_y - \sigma_x) \sin \alpha \cos \alpha$$

$$\tau = \frac{\sigma_y - \sigma_x}{2} \sin 2\alpha \quad \text{--- B*}$$

$$\sigma = \sigma_x \left(\frac{1 + \cos 2\alpha}{2} \right) + \sigma_y \left(\frac{1 - \cos 2\alpha}{2} \right)$$

$$\sigma = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\alpha \quad \text{--- B1}$$

$$\sigma - \left(\frac{\sigma_x + \sigma_y}{2} \right) = \frac{\sigma_x - \sigma_y}{2} \cos 2\alpha \quad \text{--- B2}$$

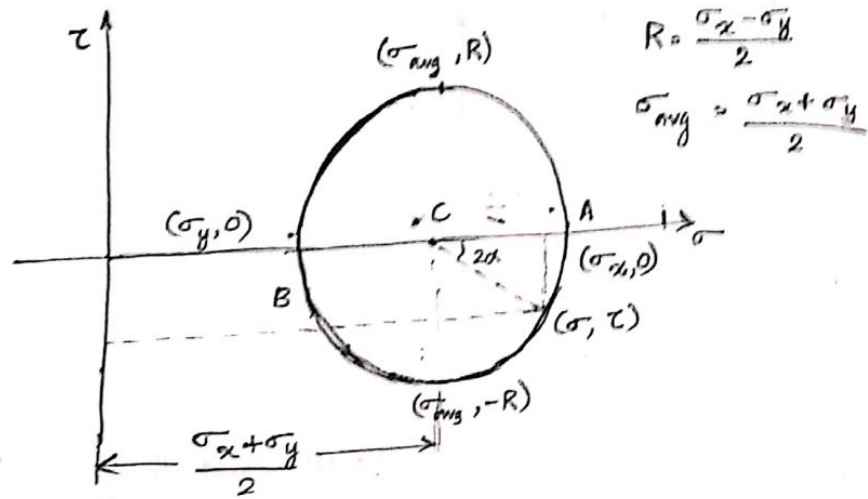
$$\tau = \frac{\sigma_y - \sigma_x}{2} \sin 2\alpha \quad \text{--- B3}$$

$$\left[\sigma - \left(\frac{\sigma_x + \sigma_y}{2} \right) \right]^2 = \left(\frac{\sigma_x - \sigma_y}{2} \cos 2\alpha \right)^2 \quad \text{--- B4}$$

$$\tau^2 = \left(\frac{\sigma_y - \sigma_x}{2} \sin 2\alpha \right)^2 \quad \text{--- B5}$$

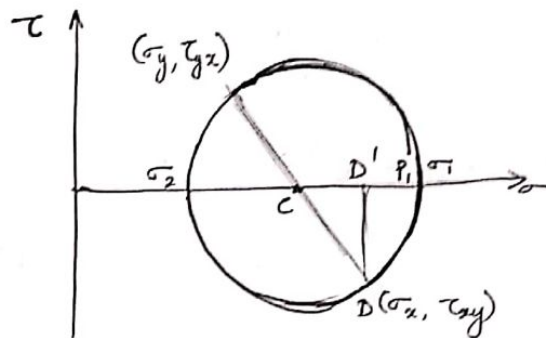
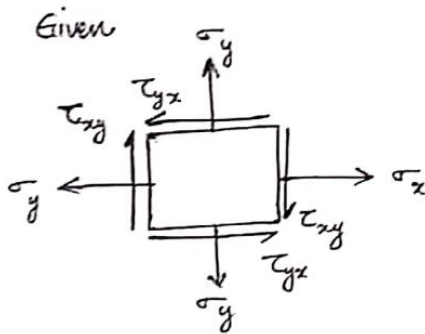
$$\Rightarrow \left(\sigma - \frac{\sigma_x + \sigma_y}{2} \right)^2 + \tau^2 = \left(\frac{\sigma_x - \sigma_y}{2} \right)^2$$

Mohr Circle



$$\sigma = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\alpha \quad - (B)$$

$$\tau = -\frac{(\sigma_x - \sigma_y)}{2} \sin 2\alpha$$



$$CD' = \sigma_x - \frac{\sigma_x + \sigma_y}{2} = \frac{\sigma_x - \sigma_y}{2}$$

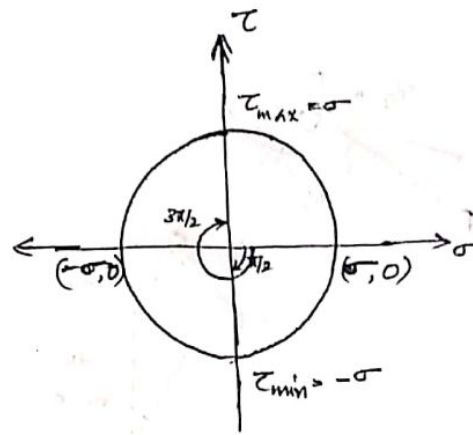
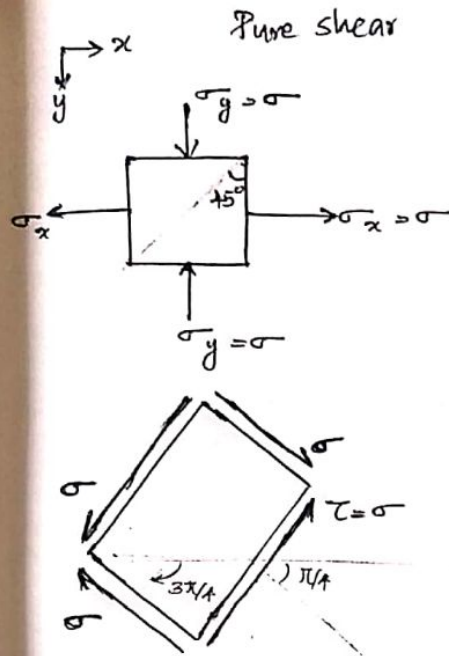
$$DD' = \tau_{xy}$$

$$CD = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + (\tau_{xy})^2}$$

$$CP_1 = CD$$

$$\sigma_1 = OC + CP_1, \quad \sigma_2 = OC - CP_1$$

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$



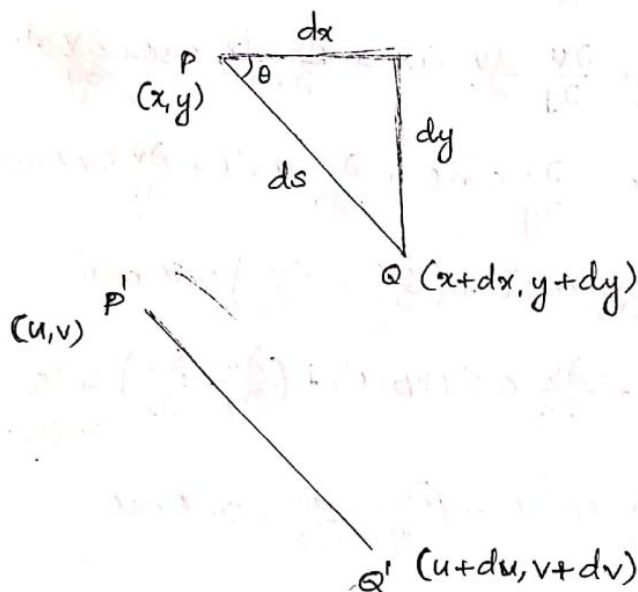
If the shear stress is positive, ~~consider~~ it causes a counter clockwise moment. and vice versa.

State of Strain at a Point

$$\epsilon_x = \frac{\partial u}{\partial x} \quad \gamma_{xy} = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}$$

$$\epsilon_y = \frac{\partial v}{\partial y} \quad \gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}$$

$$\epsilon_z = \frac{\partial w}{\partial z} \quad \gamma_{xz} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}$$



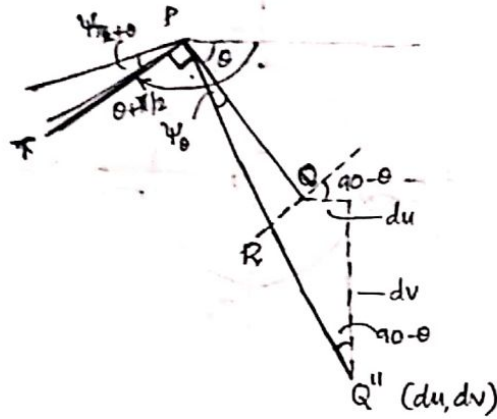
$$u = u(x, y)$$

$$v = v(x, y)$$

$$du = du(x, y)$$

$$= \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

$$dv = \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy$$



$$RQ'' = du \cos \theta + dv \cos(90-\theta)$$

$$RQ' = du \cos \theta + dv \sin \theta$$

$$\epsilon_\theta = \frac{RQ'}{PQ} = \frac{du}{ds} \cos \theta + \frac{dv}{ds} \sin \theta$$

$$= \frac{\partial u}{\partial x} \frac{dx}{ds} \cos \theta + \frac{\partial u}{\partial y} \frac{dy}{ds} \cos \theta + \frac{\partial v}{\partial x} \frac{dx}{ds} \sin \theta + \frac{\partial v}{\partial y} \frac{dy}{ds} \sin \theta$$

$$= \frac{\partial u}{\partial x} \cos^2 \theta + \frac{\partial u}{\partial y} \sin \theta \cos \theta + \frac{\partial v}{\partial x} \cos \theta \sin \theta + \frac{\partial v}{\partial y} \sin^2 \theta$$

$\frac{dx}{ds} = \cos \theta$
 $\frac{dy}{ds} = \sin \theta$

$$\epsilon_\theta = \frac{\partial u}{\partial x} \cos^2 \theta + \frac{\partial v}{\partial y} \sin^2 \theta + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \sin \theta \cos \theta$$

$$\rightarrow \epsilon_\theta = \epsilon_x \cos^2 \theta + \epsilon_y \sin^2 \theta + \gamma_{xy} \sin \theta \cos \theta$$

$$\psi_\theta PQ = QR$$

$$\psi_\theta = \frac{QR}{PQ} = \frac{-du \sin \theta + dv \sin(90-\theta)}{ds} = \frac{-du \sin \theta + dv \cos \theta}{ds}$$

$$= -\frac{\partial u}{\partial x} \frac{dx}{ds} \sin \theta + \frac{\partial u}{\partial y} \frac{dy}{ds} \sin \theta + \frac{\partial v}{\partial x} \frac{dx}{ds} \cos \theta + \frac{\partial v}{\partial y} \frac{dy}{ds} \cos \theta$$

$$= -\frac{\partial u}{\partial x} \sin \theta \cos \theta - \frac{\partial u}{\partial y} \sin^2 \theta + \frac{\partial v}{\partial x} \cos^2 \theta + \frac{\partial v}{\partial y} \sin \theta \cos \theta$$

$$\psi_\theta = -\frac{\partial u}{\partial y} \sin^2 \theta + \frac{\partial v}{\partial x} \cos^2 \theta + \left(\frac{\partial v}{\partial y} - \frac{\partial u}{\partial x} \right) \sin \theta \cos \theta$$

$$\psi_{\theta+\pi/2} = -\frac{\partial u}{\partial y} \sin^2(\theta+\pi/2) + \frac{\partial v}{\partial x} \cos^2(\theta+\pi/2) + \left(\frac{\partial v}{\partial y} - \frac{\partial u}{\partial x} \right) \sin(\theta+\pi/2) \cos(\theta+\pi/2)$$

$$= -\frac{\partial u}{\partial y} \cos^2 \theta + \frac{\partial v}{\partial x} \sin^2 \theta - \left(\frac{\partial v}{\partial y} - \frac{\partial u}{\partial x} \right) \sin \theta \cos \theta$$

$$\psi_\theta = \angle T''PQ'' - \angle T''PQ$$

$$\psi_\theta = \angle T''PQ'' - \angle T''PQ$$

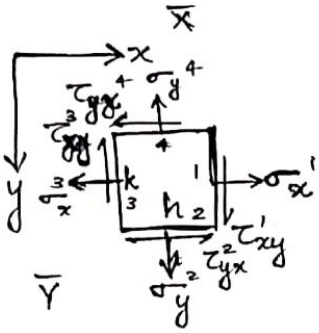
$$\gamma_\theta = \psi_\theta - \psi_{\theta'} + \pi/2$$

$$= -\frac{\partial u}{\partial y} (\sin^2 \theta - \cos^2 \theta) + \frac{\partial v}{\partial x} (\cos^2 \theta - \sin^2 \theta) + \left(\frac{\partial v}{\partial y} - \frac{\partial u}{\partial x} \right) \times 2 \sin \theta \cos \theta$$

$$= \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) (\cos^2 \theta - \sin^2 \theta) + \left(\frac{\partial v}{\partial y} - \frac{\partial u}{\partial x} \right) 2 \sin \theta \cos \theta$$

$$\rightarrow \gamma_{\theta} = \gamma_{xy} (\cos^2 \theta - \sin^2 \theta) + (\epsilon_y - \epsilon_x) 2 \sin \theta \cos \theta$$

$$\gamma_{\theta/2} = \gamma_{xy/2} (\cos^2 \theta - \sin^2 \theta) + (\epsilon_y - \epsilon_x) \sin \theta \cos \theta$$



Body forces

X - Force/volume in x direction

$\gamma_x = \text{Force/volume in } y \text{ direction}$

$$\Sigma F_x = 0$$

$$\sigma_x^1 k - \sigma_x^3 k + \tau_{xy}^2 b - \tau_{xy}^4 k + x h k = 0$$

dividing by hk

$$\frac{\sigma_x^1 - \sigma_x^3}{h} + \frac{\tau_{xy}^2 - \tau_{xy}^4}{k} + x = 0$$

$$\lim_{\substack{h \rightarrow 0 \\ k \rightarrow 0}} \frac{\sigma^1 - \sigma^3}{h} + \frac{\tau_{xy}^2 - \tau_{xy}^4}{k} + X = 0$$

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + X = 0$$

$$\Sigma F_y = 0$$

$$\sigma_{yh}^2 - \sigma_{yh}^4 + \tau_{xy}^1 k - \tau_{xy}^3 k + Yhk = 0$$

$$\frac{\sigma_y^2 - \sigma_y^4}{k} + \frac{\tau_{xy}^1 - \tau_{xy}^3}{h} + Y = 0$$

$$\lim_{\substack{h \rightarrow 0 \\ k \rightarrow 0}} \frac{\sigma_y^2 - \sigma_y^4}{k} + \frac{\tau_{xy}^1 - \tau_{xy}^3}{h} + \gamma = 0$$

$$\frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + Y = 0$$

Second order tensor

$$\tau = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix}$$

$$\begin{bmatrix} \partial/\partial x & \partial/\partial y \\ \partial/\partial z \end{bmatrix} \begin{bmatrix} \sigma_x & \tau_{xy} \\ \tau_{yx} & \sigma_y \end{bmatrix} = \begin{bmatrix} \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} & \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} \end{bmatrix}$$

$$\nabla \tau + F = 0 \quad F - \text{Body forces}$$

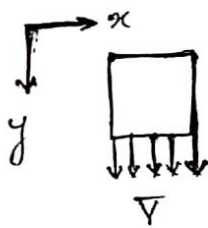
In 3 Dimension

$$x\text{-direc} \quad \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} + X = 0$$

$$y\text{-direc} \quad \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} + Y = 0$$

$$z\text{-direc} \quad \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} + Z = 0$$

$$\begin{bmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{bmatrix} \begin{bmatrix} \tau \\ \tau \\ \tau \end{bmatrix} + \begin{bmatrix} X & Y & Z \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$



$$\bar{X} = \sigma_x l + \tau_{xy} m$$

$$\bar{Y} = \tau_{yx} l + \sigma_y m$$

$$l = \cos N_x = \cos 90^\circ = 0$$

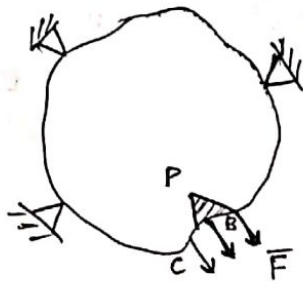
$$m = \cos N_y = \cos 0^\circ = 1$$

Boundary conditions

$$\bar{X} = \tau_{xy}$$

$$\bar{Y} = \sigma_y$$

Boundary condition



Surface load is called traction

$$\bar{X} = \sigma_x + \tau_{xy} m$$

$$\bar{Y} = \tau_{xy} + \sigma_y m - B$$

Equilibrium Equations

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + X = 0 \quad -A$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + Y = 0$$

$$\epsilon_x = \frac{\partial u}{\partial x} \quad \epsilon_y = \frac{\partial v}{\partial y} \quad \gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$$

$$\frac{\partial^2 \epsilon_x}{\partial y^2} = \frac{\partial^3 u}{\partial x \partial y^2}$$

$$\frac{\partial^2 \epsilon_y}{\partial x^2} = \frac{\partial^3 v}{\partial y \partial x^2}$$

$$\frac{\partial^2 \gamma_{xy}}{\partial x \partial y} = \frac{\partial^3 u}{\partial x \partial y^2} + \frac{\partial^3 v}{\partial y \partial x^2}$$

For u and v to be continuous it should satisfy the compatibility equation.

$$\rightarrow \frac{\partial^2 \gamma_{xy}}{\partial x \partial y} = \frac{\partial^2 \epsilon_x}{\partial y^2} + \frac{\partial^2 \epsilon_y}{\partial x^2} \quad \text{Compatibility Equation} \quad \text{--- (1)}$$

$$\epsilon_x = \frac{1}{E} (\sigma_x - \nu \sigma_y)$$

$$\epsilon_y = \frac{1}{E} (\sigma_y - \nu \sigma_x)$$

$$\gamma_{xy} = \frac{\tau_{xy}}{G} = \frac{2(1+\nu)}{E} \tau_{xy} \quad \text{--- (2)}$$

Substitute (2) in (1)

$$\frac{\partial^2 (2(1+\nu) \tau_{xy} / E)}{\partial x \partial y} = \frac{\partial^2 (\sigma_x - \nu \sigma_y) / E}{\partial y^2} + \frac{\partial^2 (\sigma_y - \nu \sigma_x) / E}{\partial x^2}$$

$$\frac{\partial^2 2(1+\nu) \tau_{xy}}{\partial x \partial y} = \frac{\partial^2 (\sigma_x - \nu \sigma_y)}{\partial y^2} + \frac{\partial^2 (\sigma_y - \nu \sigma_x)}{\partial x^2}$$

$$2(1+\nu) \frac{\partial^2 \tau_{xy}}{\partial x \partial y} = \frac{\partial^2 \sigma_x}{\partial y^2} + \frac{\partial^2 \sigma_y}{\partial x^2} - \nu \left(\frac{\partial^2 \sigma_x}{\partial y^2} + \frac{\partial^2 \sigma_y}{\partial x^2} \right) \quad \text{--- *}$$

Equilibrium equation

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + X = 0$$

diff w.r.t. x

$$\frac{\partial^2 \sigma_x}{\partial x^2} + \frac{\partial^2 \tau_{xy}}{\partial x \partial y} + \frac{\partial X}{\partial x} = 0 \quad \text{--- 3a}$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + Y = 0$$

diff. w.r.t. y

$$\frac{\partial^2 \tau_{xy}}{\partial x \partial y} + \frac{\partial^2 \sigma_y}{\partial y^2} + \frac{\partial Y}{\partial y} = 0 \quad \text{--- 3b}$$

Adding 3a and 3b

$$\frac{\partial^2 \sigma_x}{\partial x^2} + \frac{\partial^2 \sigma_y}{\partial y^2} = -2 \frac{\partial^2 \tau_{xy}}{\partial x \partial y} - \frac{\partial X}{\partial x} + \frac{\partial Y}{\partial y} \quad \text{--- ④}$$

*

$$\frac{\partial^2 \sigma_x}{\partial y^2} + \frac{\partial^2 \sigma_y}{\partial x^2} - \nu \left(\frac{\partial^2 \sigma_x}{\partial y^2} + \frac{\partial^2 \sigma_y}{\partial x^2} \right) = -(1+\nu) \left[\frac{\partial^2 \sigma_x}{\partial x^2} + \frac{\partial^2 \sigma_y}{\partial y^2} + \frac{\partial X}{\partial x} + \frac{\partial Y}{\partial y} \right]$$

$$\frac{\partial^2 \sigma_x}{\partial y^2} + \frac{\partial^2 \sigma_y}{\partial x^2} = - \left[\frac{\partial^2 \sigma_x}{\partial x^2} + \frac{\partial^2 \sigma_y}{\partial y^2} \right] - (1+\nu) \left[\frac{\partial X}{\partial x} + \frac{\partial Y}{\partial y} \right]$$

$$\frac{\partial^2 \sigma_x}{\partial y^2} + \frac{\partial^2 \sigma_y}{\partial x^2} + \frac{\partial^2 \sigma_x}{\partial x^2} + \frac{\partial^2 \sigma_y}{\partial y^2} = -(1+\nu) \left[\frac{\partial X}{\partial x} + \frac{\partial Y}{\partial y} \right]$$

$$\rightarrow \left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right] (\sigma_x + \sigma_y) = -(1+\nu) \left[\frac{\partial X}{\partial x} + \frac{\partial Y}{\partial y} \right] \quad \text{--- ⑤}$$

For plane stress

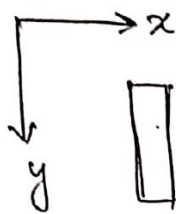
Stress Function

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + X = 0 \quad \text{--- *}$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + Y = 0$$

$$\frac{\partial^2 \epsilon_x}{\partial y^2} + \frac{\partial^2 \epsilon_y}{\partial x^2} = \frac{\partial^2 \gamma_{xy}}{\partial x \partial y}$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) (\sigma_x + \sigma_y) = -(1+\nu) \left(\frac{\partial X}{\partial x} + \frac{\partial Y}{\partial y} \right) \quad \text{--- Plane stress}$$



$$x=0$$

$$Y = \rho g$$

Airy's Experiment

$$\left. \begin{aligned} \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} &= 0 \\ \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \rho g &= 0 \end{aligned} \right\} \text{--- (a)}$$

$$\left. \begin{aligned} \sigma_x &= \frac{\partial^2 \phi}{\partial y^2} - \rho g y \\ \sigma_y &= \frac{\partial^2 \phi}{\partial x^2} - \rho g y \\ \tau_{xy} &= -\frac{\partial^2 \phi}{\partial x \partial y} \end{aligned} \right\} \text{--- (b)}$$

Substitute (b) in (a)

$$\frac{\partial^3 \phi}{\partial x \partial y^2} - \frac{\partial^3 \phi}{\partial x \partial y^2} = 0$$

$$-\frac{\partial^3 \phi}{\partial x^2 \partial y} + \left(\frac{\partial^3 \phi}{\partial x^2 \partial y} - \rho g \right) + \rho g = 0$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)(\sigma_x + \sigma_y) = 0$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)\left(\frac{\partial^2 \phi}{\partial y^2} - \rho g y + \frac{\partial^2 \phi}{\partial x^2} - \rho g y\right) = 0$$

$$\frac{\partial^2 \phi}{\partial x^4} + \frac{\partial^2 \phi}{\partial y^4} + 2 \frac{\partial^2 \phi}{\partial x^2 \partial y^2} = 0$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)\phi = 0$$

$$\nabla^2 \nabla^2 \phi = 0$$

→ General case $\nabla^4 \phi(x, y) = 0$ Biharmonic

$$F = -\nabla V(x, y) \Rightarrow V = \text{Body force potential}$$

$$\Rightarrow X = -\frac{\partial V}{\partial x}$$

$$Y = -\frac{\partial V}{\partial y}$$

$$\Rightarrow \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{-\partial V}{\partial x} = 0$$

$$\frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{-\partial V}{\partial y} = 0$$

$$\left. \begin{aligned} \frac{\partial}{\partial x}(\sigma_x - V) + \frac{\partial \tau_{xy}}{\partial y} &= 0 \\ \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial}{\partial y}(\sigma_y - V) &= 0 \end{aligned} \right\} - a$$

$$\left. \begin{aligned} \sigma_x - V &= \frac{\partial^2 \phi}{\partial x^2} \\ \sigma_y - V &= \frac{\partial^2 \phi}{\partial y^2} \\ \tau_{xy} &= -\frac{\partial^2 \phi}{\partial x \partial y} \end{aligned} \right\} - b$$

⑥ in ① identically satisfies

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) \left[\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \partial v(x, y) \right]$$

$$= (1+\nu) \left[\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right]$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) \phi = (\nu-1) \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) v$$

$$\nabla^4 \phi(x, y) = (\nu-1) \nabla^2 v(x, y)$$